Supplemental Material: Hedging against prediction error through inference delay in world-model planning

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Supplemental Notes 1–2 summarize the decision rules and compute reporting for the studies in the main paper, including the further tasks, external models, hedge-then-refine and real-time evaluations. The full study plan and its amendments are in `paper/docs/EXPERIMENT_PLAN_SIDE.md` in the public code release at `https://github.com/rxailab/residual-calibrated-hedging`.

SUPPLEMENTAL NOTE 1: PRE-SPECIFIED HYPOTHESES AND DECISION RULES

The dates below identify when the corresponding decision rules were specified. Pilot overlap, amendments and follow-ups are stated explicitly; pre-specification does not imply that every pooled sample was untouched when its contrast was chosen. The full study design is `EXPERIMENT_PLAN_SIDE.md` in the code release.

- **Study 1 (4 October 2026).** H1: committed-prefix minus time-aligned return > 0 at each of $d = 2, 4, 8$, pooled over cells above the 0.15 floor. Result: $p = 0.050$, not supported. H2: committed-prefix removes at least half of the time-aligned mismatch loss at $d = 4, 8$. Result: not supported (27–31%).
- **Hedging pilot (5 October 2026, before its runs).** Contrast $R(K = 16) - R(K = 1)$ at $d = 2, 4, 8$ on one model per cell. Rule: proceed if the 95% interval lies above 0 at two or more delays; fall back if it includes 0 at all three. Result: proceed ($d = 8$ and, marginally, $d = 4$).
- **Replication (5 October 2026, before its runs).** Over 24 models, $R(K = 16) - R(K = 1)$ at $d = 4$ and $d = 8$, one-sided, Holm-corrected. Secondary: $d = 2$, and $K = 4$ against $K = 1$. Result: supported at $d = 8$ (Holm $p = 0.0014$), not at $d = 4$ ($p = 0.19$).
- **Equal latency (5 October 2026, before its runs).** $R(K = 4, N = 512) - R(K = 1, N = 512)$ and $R(K = 16, N = 128) - R(K = 1, N = 512)$ at $d = 4, 8$, Holm over the four contrasts. Result: $K = 4$ supported at both delays (Holm $p = 0.035$, 0.0004); $K = 16$ at $N = 128$ worse at both. The $K = 4$ contrast had been secondary in the replication plan; it was made confirmatory here after the pilot, whose eight models it includes, had been run (on the 16 replication models alone: +0.025, +0.037).
- **Held-out method study (6 October 2026, before its runs).** Tuning of the hand-off and risk penalty strengths on the seed-0 models with separate episodes, by a fixed rule (chosen: $\lambda_h = 0.25$, $\lambda_r = 0.5$; the hand-off penalty lowered tuning return at every strength). Primary contrasts on the 16 seed-1/seed-2 models at $d = 4, 8$, Holm over eight:
  - C1: learned error model vs residual bank. Result: not supported (+0.001, +0.011).
  - C2: hand-off penalty vs none. Result: not supported; the penalty lowered return (−0.026, −0.021).
  - C3: hand-off vs risk penalty. Result: not supported (−0.011, +0.006).
  - C4: full variant (learned model with hand-off penalty) vs committed prefix. Result: not supported (−0.001, +0.026).

  Secondary: both hedges beat committed-prefix planning (+0.025/+0.037 for the bank, +0.025/+0.047 for the learned model, at $d = 4/8$).
- **Ensemble and particle baselines (11 October 2026, before their runs).** RCH ($K = 4$) against a three-model ensemble at matched population and at matched latency, $d = 4, 8$, two-sided, Holm over four. Results: RCH below the latency-matched ensemble at $d = 4$ (−0.042, Holm $p = 0.0008$); the other three differences favored the ensemble with intervals including zero. Secondary: the ensemble against committed-prefix planning, the ensemble with RCH, and particle anchors (see the ensemble comparison in the main paper). Two post-hoc checks followed, each written into the plan before its runs: RCH at 2 iterations, and the paused ensemble (see the main paper's ensemble-baseline Methods).
- **Longer delay (16 October 2026, before its runs).** $R(K = 4) - R(K = 1)$ at $d = 16$ over the 24 models, one-sided, a single test. Result: supported (+0.023, $p = 0.0094$).

- **Real-time hedging (16 October 2026, before its runs).** Real-time return of hedged minus committed-prefix planning on the six cup-catch models, one-sided. Result: not supported ($+0.074$ $[-0.029, +0.181]$, $p = 0.082$).
- **Real-time replication (16 October 2026, before its runs).** The same test on 100 new episodes per arm and model. Result: not supported ($+0.071$ $[-0.013, +0.150]$, $p = 0.051$).
- **Third-party model (16 October 2026, amended twice before the confirmatory runs).** Success of hedged minus committed-prefix planning with LeWM in the hand-over setting, pooled over three tasks, at $d = 2$ and $d = 4$, one-sided, Holm over two. Result: not supported ($+0.002$, $+0.015$; Holm $p = 0.48$, 0.34). The amendments replaced replanning every decision by LeWM's 5-decision period after a harness check, and a long-horizon setting by the hand-over setting after smoke tests showed no effect of delay there (see the main paper's third-party-model Methods).
- **Further tasks (16 October 2026, before data collection).** $R(K = 4) - R(K = 1)$ at $d = 4$ and $d = 8$ on the eligible cells of reacher easy, cheetah run and finger spin, one-sided, Holm over two. Result: supported ($+0.025$, $+0.017$; Holm $p = 0.0004$, 0.0018).
- **Offline value of hedging (16 October 2026, before the further tasks' closed-loop runs; secondary).** Positive rank correlation between the cell-mean offline value V and the closed-loop gain, one-sided. Result: supported on the original cells as a retrodiction ($+0.69$, $p = 0.002$); not supported as a prediction for the further tasks ($+0.33$, $p = 0.13$).
- **Hedge-then-refine (17 October 2026, before its runs).** $R(\text{hedge-then-refine}) - R(\text{RCH})$ at $d = 4$ and $d = 8$ over the 45 models, one-sided, Holm over two. Result: supported at $d = 8$ ($+0.015$, Holm $p = 0.0098$), not at $d = 4$ ($+0.008$, $p = 0.08$).
- **Powered real-time test (17 October 2026, before its runs).** Real-time return of hedge-then-refine minus committed-prefix planning over 21 models in the seven cells where its fixed-delay gain was at least 0.03, one-sided. Result: supported ($+0.055$ $[+0.031, +0.081]$, $p < 0.0001$). Five models ran on exclusive V100 nodes, nine on shared V100 nodes and seven on shared L40 or L40S nodes, a change made during the study to free shared GPUs; each model ran all arms under one condition with its own calibrated delay.
- **Released TD-MPC2 models (17 October 2026, before their runs).** $R(\text{RCH}) - R(\text{committed prefix})$ at $d = 4$ and $d = 8$ in the pipelined delay setting, pooled over six tasks, one-sided, Holm over two. Result: not supported ($+0.004$, -0.019; Holm $p = 0.81$).
- **Log-probability anchors (17 October 2026, after the previous result and before these runs).** The same contrast with log-probability anchors on new episodes. Result: not supported ($+0.008$, -0.030). On the 15 pixel TD-MPC2 models (secondary), log-probability and additive anchors did not differ ($+0.008$, $+0.001$).

SUPPLEMENTAL NOTE 2: COMPUTE

Reported approximate study-level estimates, in GPU-hours, are: 30 for the Study 1 and 2 return runs, 15 for the hedging pilot, 30 for replication, 8 for the equal-latency study, and 46 for diffusion evaluation. Later estimates are 13 for ensemble and particle baselines, including follow-up checks and timing; 1 for the longer-delay study; 16 on exclusive V100 nodes for real-time hedging and its replication; 12 for LeWM, including smoke tests and timing; about 45 for the further tasks, including collection, training 27 models, closed-loop evaluation and diagnostics; 4 for offline values on the original models; 9 for hedge-then-refine; and 13 for the released TD-MPC2 evaluations and follow-up anchors.

The experiment log records 13.2 GPU-hours for the ensemble-baseline study and 22.5 GPU-hours jointly for Sections 17–18 (hedge-then-refine and released TD-MPC2, including the anchor follow-up). These overlap the rounded estimates above and are not additional work. The powered real-time study was budgeted at about 21 one-GPU job-hours; a reconciled actual-use total is not recorded in the public log. Five models ran on exclusive V100 nodes, nine on shared V100 nodes and seven on shared L40 or L40S nodes. GPU-hours across devices and exclusive-node allocations are not interchangeable.

Original training belongs to a separate study; these evaluations reuse its checkpoints, with three models per ensemble. Only the 27 further-task models were newly trained here. Additional timing and implementation work is not fully itemized. No audited aggregate of training and evaluation compute is claimed.